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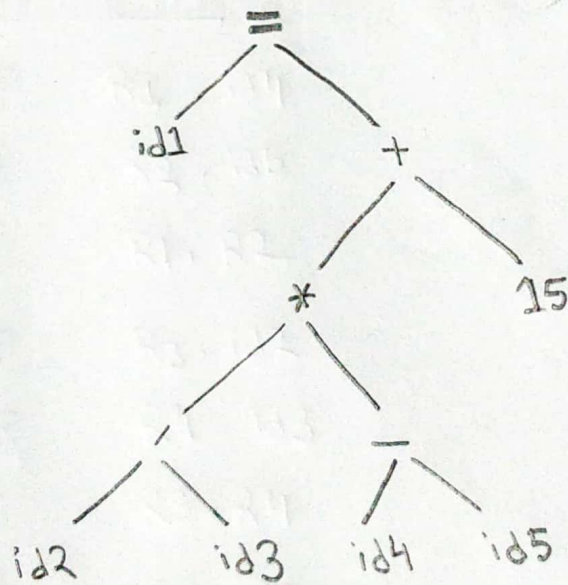
1. here,

$$p = x/y * (y-z) + 15$$

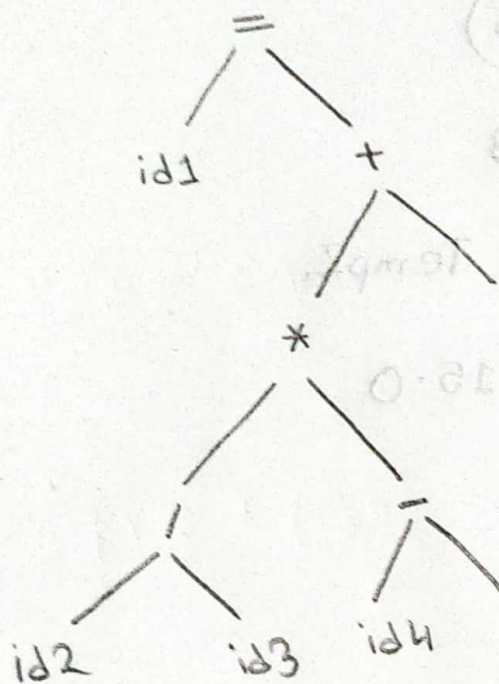
Now lexical analyzer,

$$id1 = id2/id3 * (id4-id5) + 15$$

Syntax analyzer,



Semantic analyzer,



Intermediate code generator:

Temp1 = (ID4 - ID5)

Temp2 = id2 / id3

Temp3 = Temp1 * Temp2

Temp4 = int to real (15)

Temp5 = Temp3 + Temp4

id1 = Temp5

code optimizer:

$$\text{Temp1} = (\text{id4} - \text{id5})$$

$$\text{Temp2} = \text{id2} / \text{id3}$$

$$\text{Temp3} = \text{Temp1} * \text{Temp2}$$

$$\text{id1} = \text{Temp3} + 15.0$$

code generator:

MOV F R1, id4

MOV F R2, id5

SUB F R1, R2

MOV F R3, id2

MOV F R4, id3

DIV F R3, R4

MUL F R1, R3

ADD F R1, 15.0

MOV F id1, R1

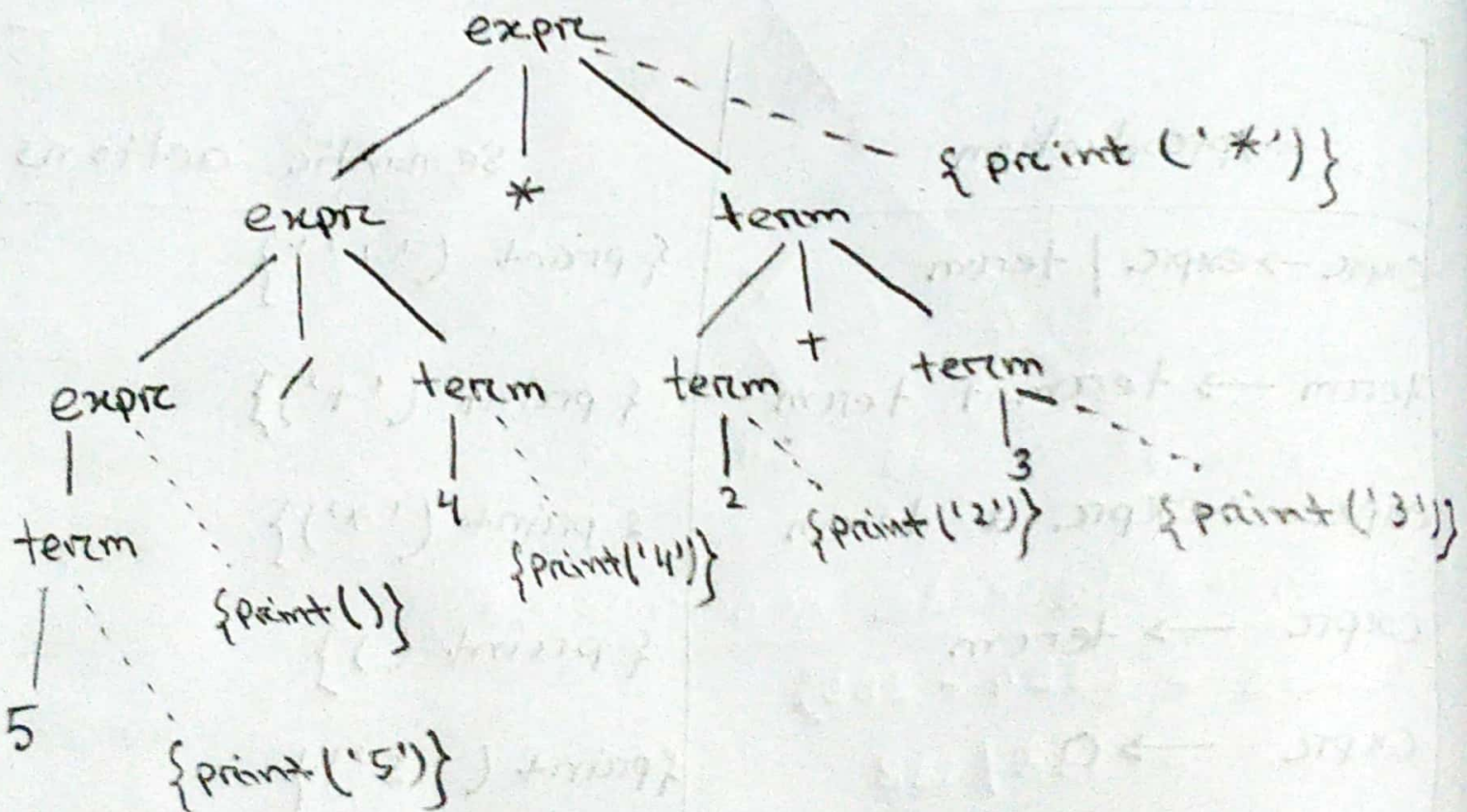
2. here the string is

$$5/4 * 2 + 3$$

production	semantic actions
$expr \rightarrow expr / term$	{ print ('/') }
$term \rightarrow term + term$	{ print ('+') }
$expr \rightarrow expr * term$	{ print ('*') }
$expr \rightarrow term$	{ print () }
$expr \rightarrow 5$	{ print ('5') }
$\vdots$	{ print ('4') }
$\vdots$	{ print ('2') }
$\vdots$	{ print ('3') }
$expr \rightarrow 9$	

infix = 5/4 * 2+3

Postfix = 5 4 / 2 3 + *



3.

production	Semantic Rules	Translation Schema
$S \rightarrow U$	$S.T = U.T$	$\{ \text{print} () \}$
$U \rightarrow T a U$	$U.t = T.t \parallel U.t \parallel 'a'$	$\{ \text{print} ('a') \}$
$U \rightarrow T a T$	$U.t = T.t \parallel T.t \parallel 'a'$	$\{ \text{print} ('a') \}$
$T \rightarrow a T b T$	$T.t = T.t \parallel T.t \parallel 'a' \parallel 'b'$	$\{ \text{print} ('a'), \text{print} ('b') \}$
$T \rightarrow b T a T$	$T.t = T.t \parallel T.t \parallel 'b' \parallel 'a'$	$\{ \text{print} ('b'), \text{print} ('a') \}$
$T \rightarrow d$	$T.t = 'd'$	$\{ \text{print} ('d') \}$